API Security Risk Analysis Report

Target API: jsonplaceholder.typicode.com

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About the Report

This report presents an API security risk analysis conducted on a public demo API.

The objective is to identify common API security issues and provide recommendations to improve security practices.

Objective

- Analyze API endpoints
- Identify security risks
- Assess authentication and data exposure
- Provide remediation recommendations

Tools Used

- Postman (API testing)
- Browser DevTools
- API documentation

API Overview

API Tested:

https://jsonplaceholder.typicode.com

Endpoints Analyzed:

- /posts
- /users
- /comments

API Type:

Public demo API for testing and learning purposes.

Finding 1: No Authentication Required

Description:

The API allows access to endpoints without any authentication.

What happens?

Users can access data without login or API key.

Risk:

If used in a real system, this could expose sensitive data to unauthorized users.

Risk Level: Medium

Recommendation:

Implement authentication using API keys, tokens, or OAuth.

Finding 2: Excessive Data Exposure

Description:

The API returns full datasets including user information.

What happens?

Users receive more data than necessary.

Risk:

Attackers can collect data for further attacks or misuse.

Risk Level: Medium

Recommendation:

Limit response data to only required fields.

Finding 3: No Rate Limiting

Description:
The API does not limit the number of requests.

What happens?
Users can send unlimited requests.

Risk:
This may lead to abuse or denial-of-service risks.

Risk Level: Medium

Recommendation:
Implement rate limiting to control request frequency.

Finding 4: Predictable Endpoints

Description:

Endpoints follow predictable patterns (e.g., /posts/1, /posts/2).

What happens?

Attackers can easily enumerate data.

Risk:

Unauthorized access to large datasets.

Risk Level: Low

Recommendation:

Implement proper access control and validation.

Risk Summary

- High: 0
- Medium: 3
- Low: 1

Recommendations

- Implement authentication mechanisms
- Limit data exposure
- Apply rate limiting
- Secure API endpoints

Conclusion

The API demonstrates common security weaknesses such as lack of authentication and excessive data exposure.

While acceptable for testing, these issues would be critical in production systems.

Implementing proper API security controls is essential.